VisInNet:
A Deep Visual-Inertial Odometry Approach

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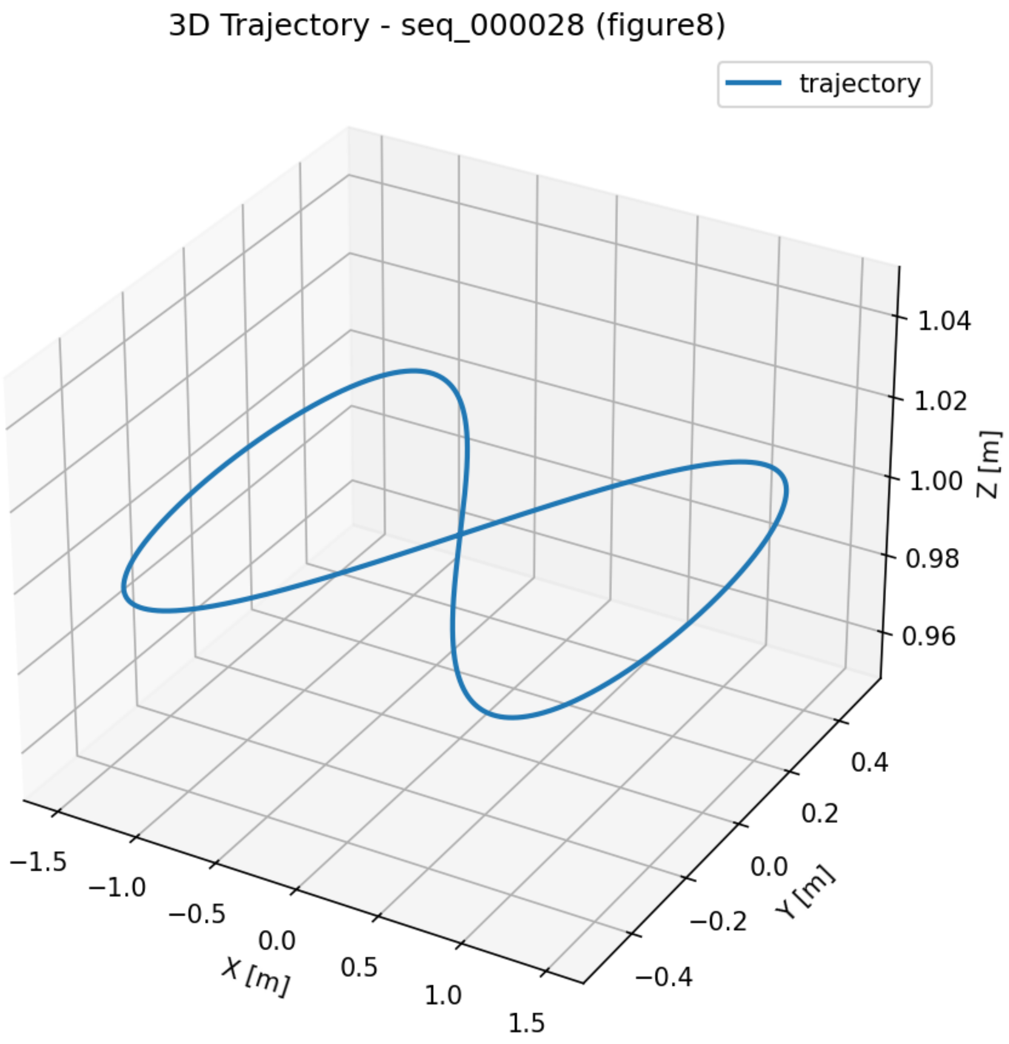
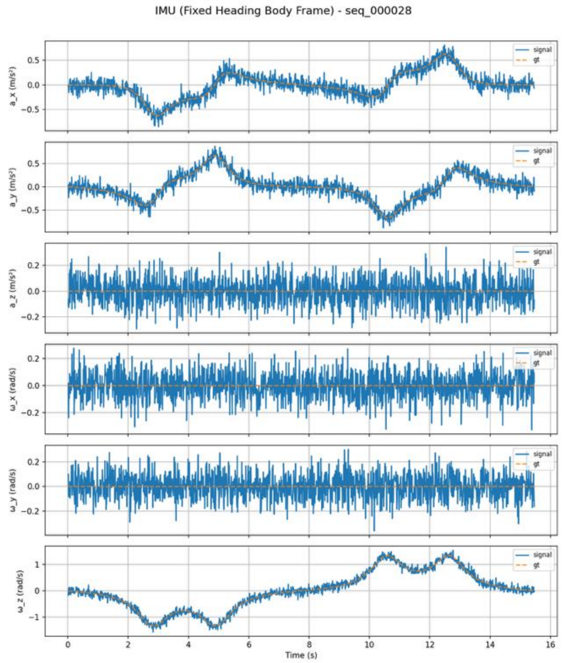
Synthetic Data Generation

Visual Data

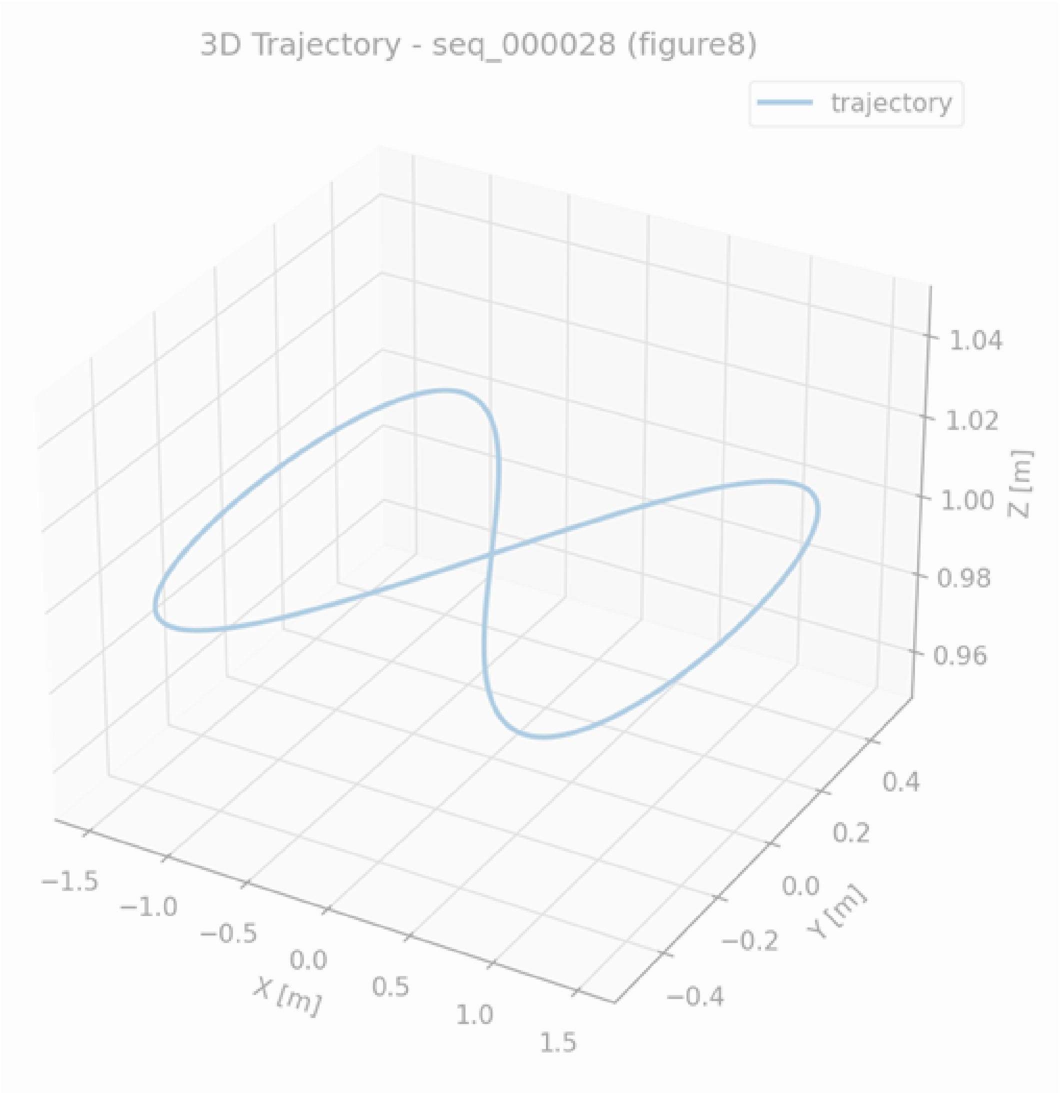


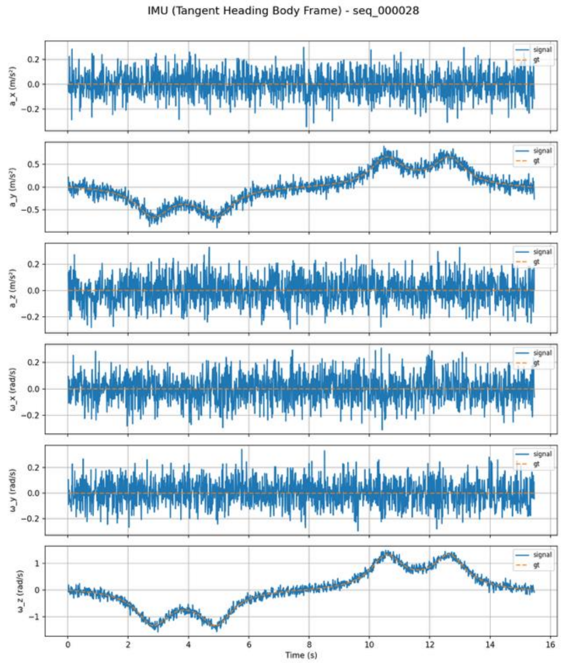


Inertial Data:
Fixed Heading

Synthetic Data Generation

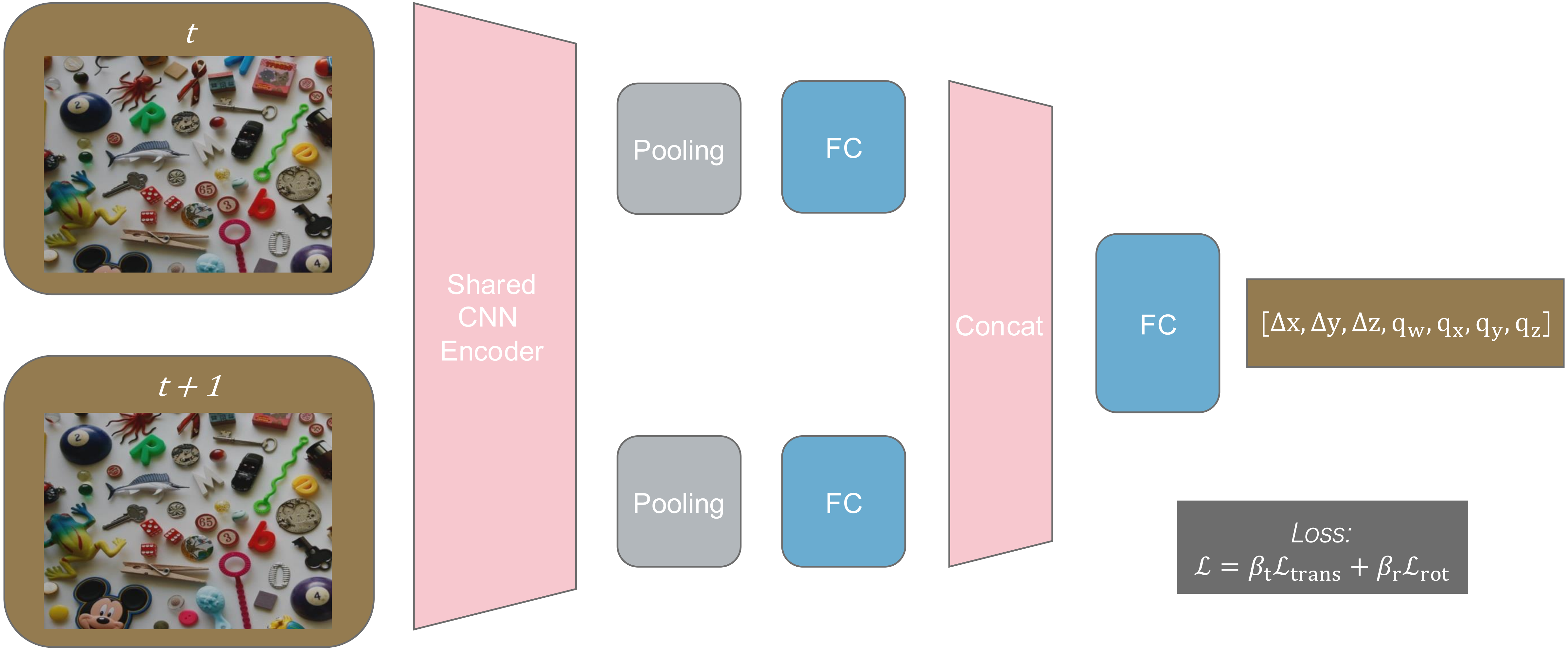
Visual Data



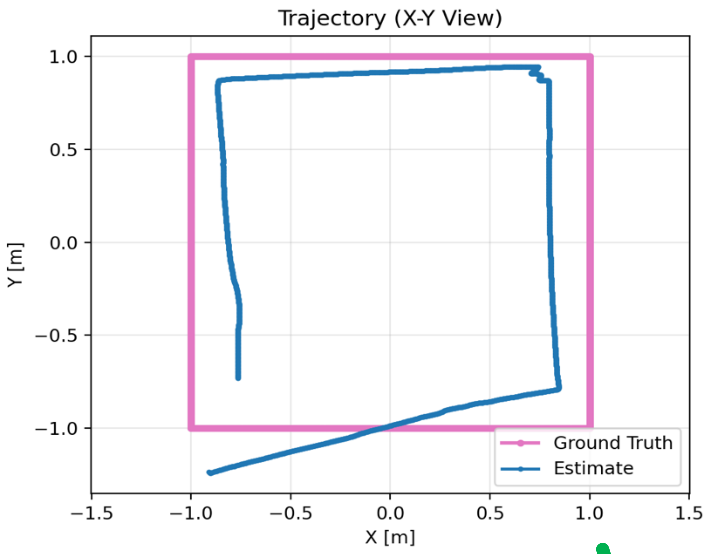


Inertial Data:
Tangent Heading

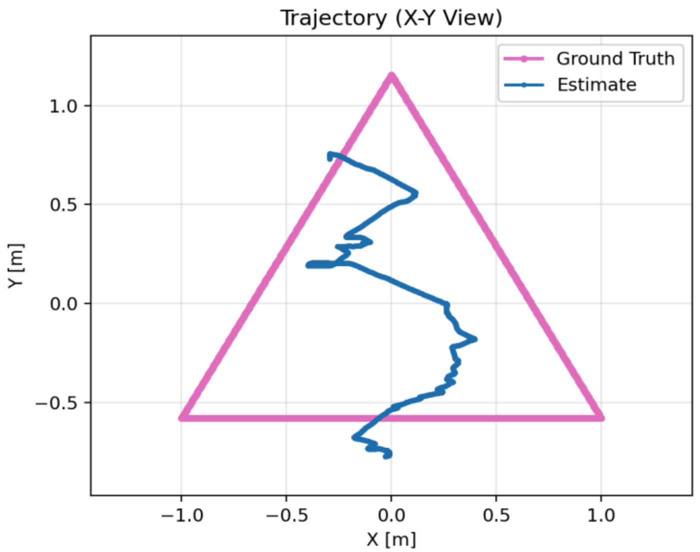
Visual Network



Visual Network: Train and Test Results



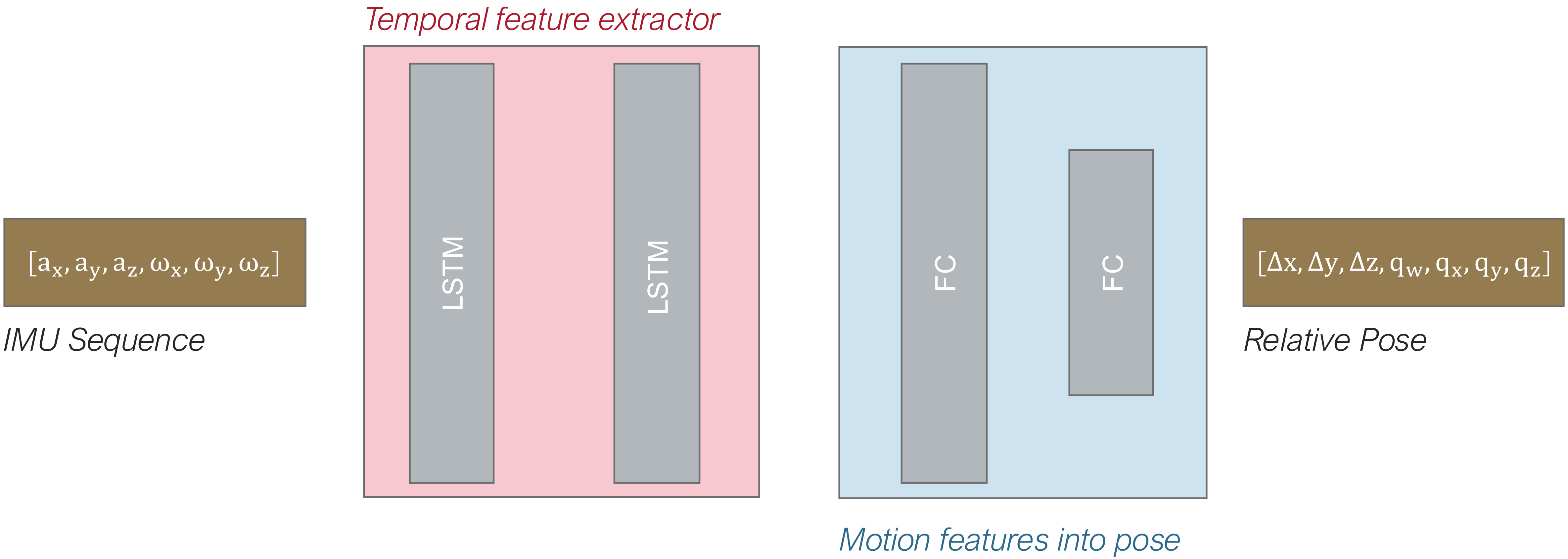
Train RMSE ATE:
0.196



Test RMSE ATE:
0.690

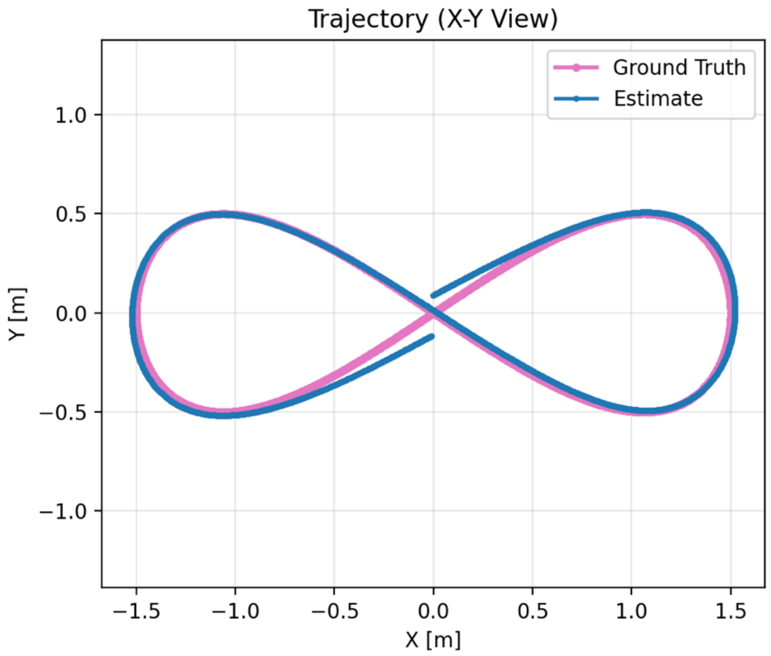
Likely cause: low variance in dataset

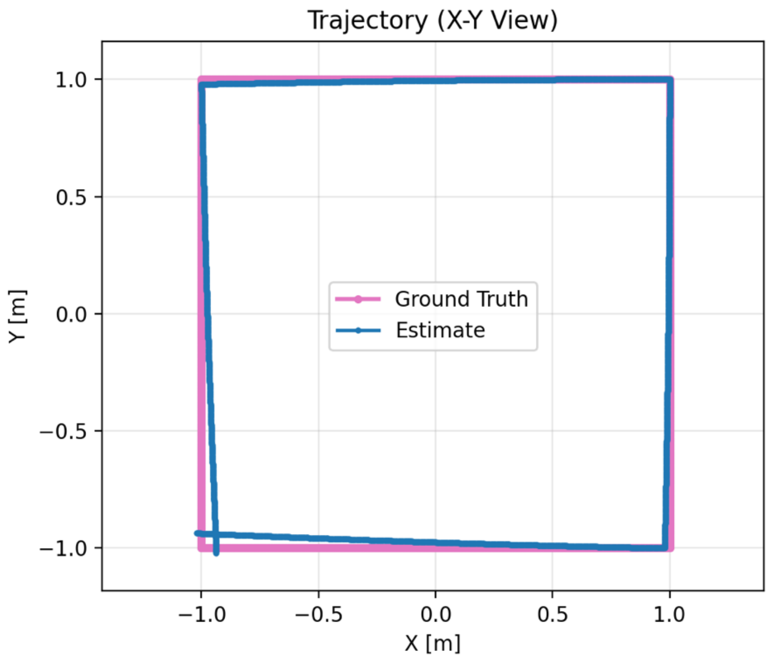
Inertial Network

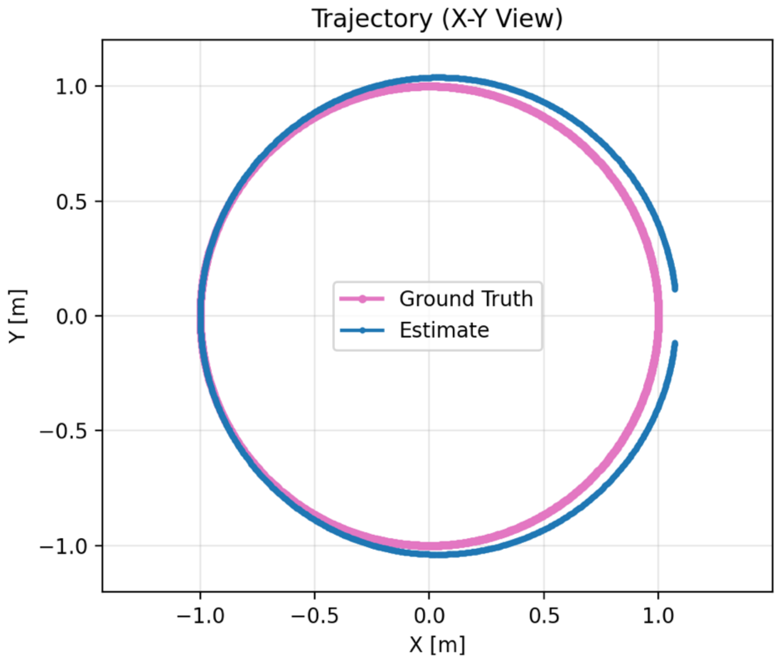


Motion depends on past measurements ---> LSTM

Inertial Network: Train Results

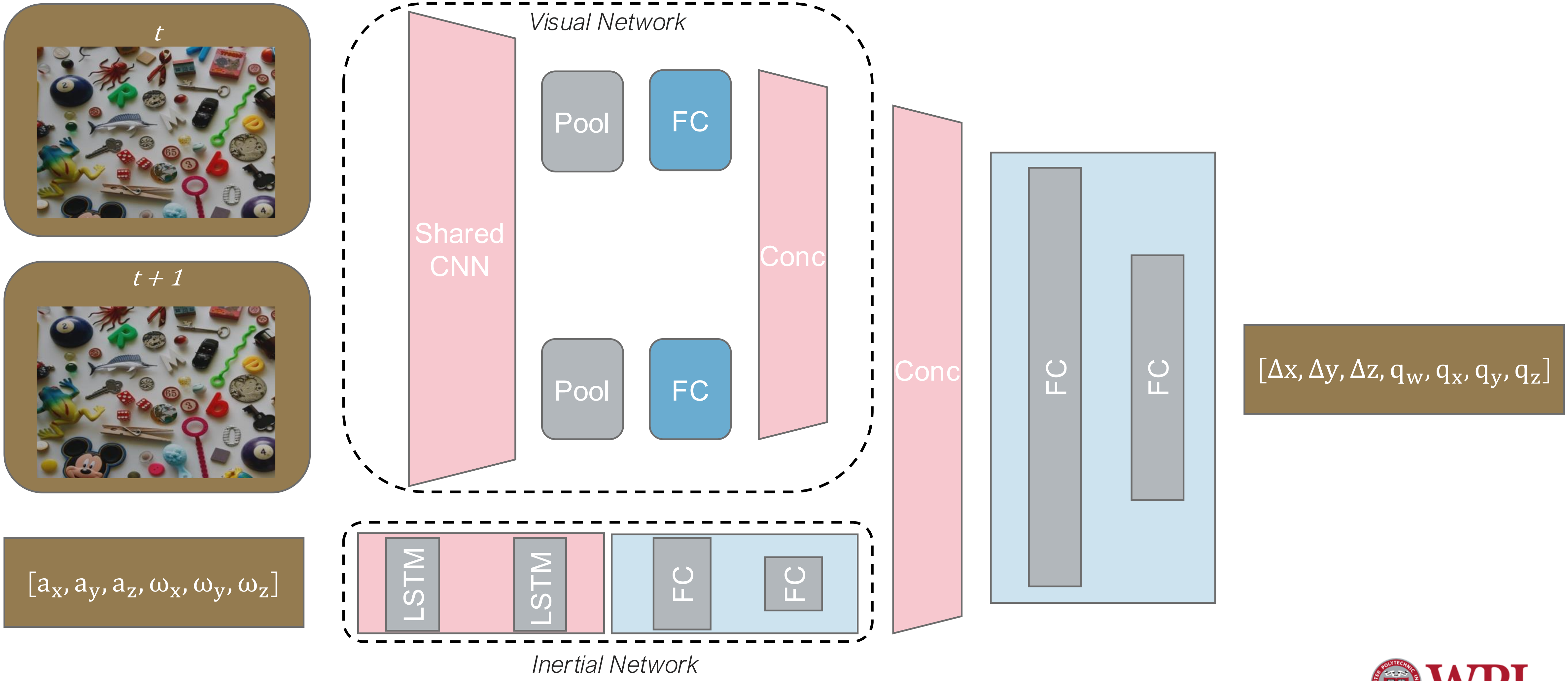




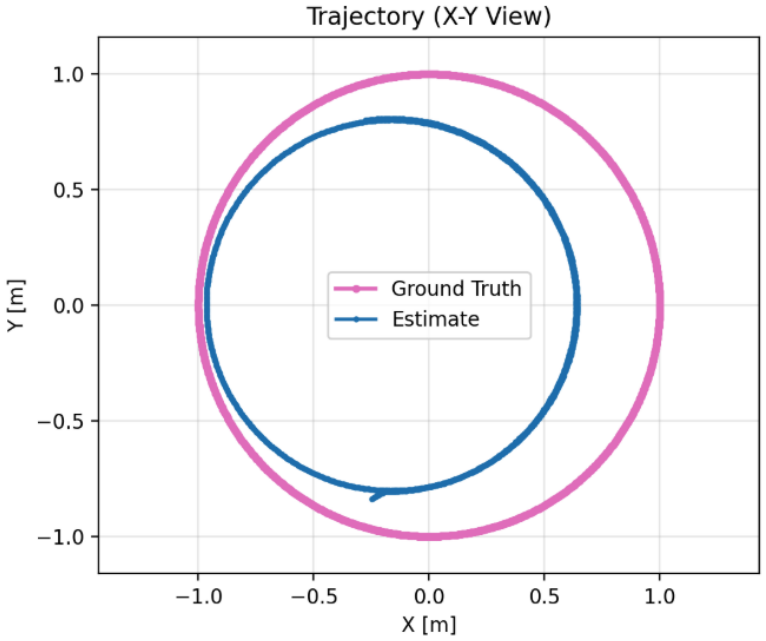


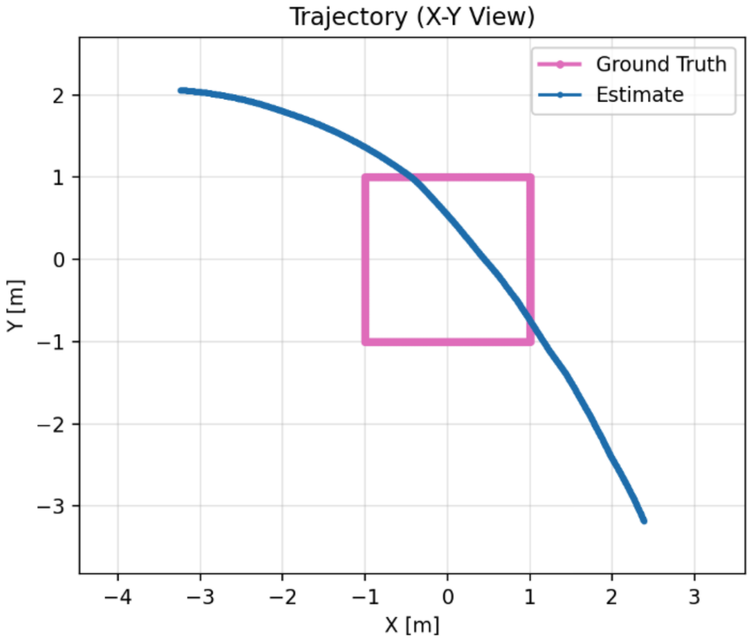
WPI

Visual-Inertial Combination



Visual-Inertial Combination: Results





Next step: new loss function

Thank you